

CLINICAL PROBLEM-SOLVING

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A “Fluid” Diagnosis

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Michele Swanson, Ph.D., and Jane C. Deng, M.D.*In this Journal feature, information about a real patient is presented in stages (boldface type) to an expert clinician, who responds to the information by sharing relevant background and reasoning with the reader (regular type). The authors' commentary follows.*

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A 61-year-old woman presented to an emergency department in Michigan after a syncopal episode. She reported a 3-week history of fatigue, malaise, and generalized weakness. Laboratory evaluation revealed new anemia with a hematocrit of 19.7% and a white-cell count of 16,100 per cubic millimeter, with 39% blast cells in peripheral blood. She received a diagnosis of acute myeloid leukemia, was transferred to a tertiary care hospital, and began therapy with standard-dose cytarabine and daunorubicin. Her hospital course was uneventful until hospital day 10, when a fever developed, with a temperature of 38.7°C.

Patients with acute myeloid leukemia have dysfunction of myeloid cells — notably neutrophils — which is exacerbated by chemotherapy. I would therefore be concerned that the fever represents infection by a pathogen for which neutrophil responses are important, specifically in cases of bacterial and fungal (e.g., aspergillus and candida) infections.

The patient reported no history of headache, visual changes, sinus or tooth discomfort, sore throat, chest pain, cough, dyspnea, abdominal pain, nausea, emesis, diarrhea, back pain, dysuria, rash, or new skin lesions. Her medical history included type 2 diabetes mellitus, hypertension, hyperlipidemia, nephrolithiasis, and depression. Her medications at the time of evaluation included insulin, pravastatin, sertraline, and trazodone. She was receiving allopurinol for prophylaxis against tumor lysis syndrome, as well as acyclovir and voriconazole for antimicrobial prophylaxis. She had no known drug allergies, and her family history was notable for a mother with lung cancer. She reported being a current smoker and had a smoking history of 15 pack-years; she reported no history of alcohol or recreational drug use. She worked in the laundry and custodial services at a skilled nursing facility. She had visited a casino in the week before her hospitalization but reported no history of travel outside of the state of Michigan. She was in a sexually monogamous relationship and reported no history of sexually transmitted infections.

Smoking results in mucosal alterations that promote colonization by potentially pathogenic bacteria and impair innate and adaptive immune-cell function. Smokers consequently have a dose-dependent increased risk of community-acquired pneumonia, particularly pneumonia that is caused by pneumococcus or atypical pathogens such as legionella. Smokers also have an increased risk of influenza, other community-acquired respiratory viral infections, and varicella pneumonitis.

The patient's exposure history suggests a possible predisposition to other infections with causes such as *Histoplasma capsulatum*, which is endemic in Michigan, or drug-resistant nosocomial pathogens, which may colonize nursing home workers and lead to invasive infections in patients receiving chemotherapy.

The patient's temperature was 38.7°C, heart rate 83 beats per minute, respiratory rate 18 breaths per minute, blood pressure 120/55 mm Hg, and oxygen saturation 97% while she was breathing ambient air. The patient's appearance was without evidence of systemic toxicity, and she had no oropharyngeal lesions, sinus tenderness, or cervical lymphadenopathy. Auscultation of the heart revealed a regular rate and rhythm without murmurs, rubs, or gallops. The lungs were clear on auscultation. Results from an abdominal examination were benign. A skin examination revealed no abnormalities. She had a peripherally inserted central catheter (PICC) but no other indwelling lines or devices.

The patient's white-cell count was 300 per cubic millimeter, with 46.0% neutrophils, 48.0% lymphocytes, 1.0% monocytes, 4.0% eosinophils, and 0% basophils. The hematocrit was 24.2%, and the platelet count 34,000 per cubic millimeter. Findings from kidney- and liver-function tests were unremarkable. A nasopharyngeal polymerase-chain-reaction (PCR) test was negative for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). A urinalysis revealed no abnormalities, and a chest radiograph was unremarkable.

This patient has febrile neutropenia, which is defined as the presence of fever and an absolute neutrophil count of less than 1500 per cubic millimeter. Given the absence of focal symptoms, I would start empirical intravenous therapy with broad-spectrum antibiotics with antipseudomonal activity. In the absence of findings suggestive of invasive fungal infection, I would not escalate antifungal treatment unless her fever persists beyond 72 hours. Two sets of blood cultures should be obtained, and because she has a PICC, one of the two sets should be obtained from this line.

The patient began intravenous therapy with cefepime. Blood and urine cultures, obtained before the initiation of empirical antibiotic therapy, re-

mained without growth; one set of blood cultures was obtained from the PICC. The patient remained persistently febrile, and tachypnea developed and worsened, without concurrent hypoxemia, tachycardia, or hypotension. On hospital day 13, chest radiography was repeated and revealed a new, dense consolidation in the left upper lobe (Fig. 1). Computed tomography of the chest revealed a masslike consolidative lesion in the left upper lobe with surrounding ground-glass opacifications and mildly enlarged ipsilateral mediastinal lymph nodes (Fig. 2).

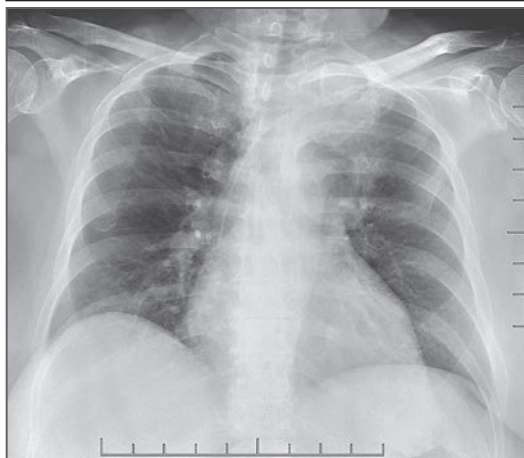


Figure 1. Chest Radiograph Obtained on Day 13 of Hospitalization.

A chest radiograph obtained on day 13 of hospitalization shows a dense consolidation in the left upper lobe.

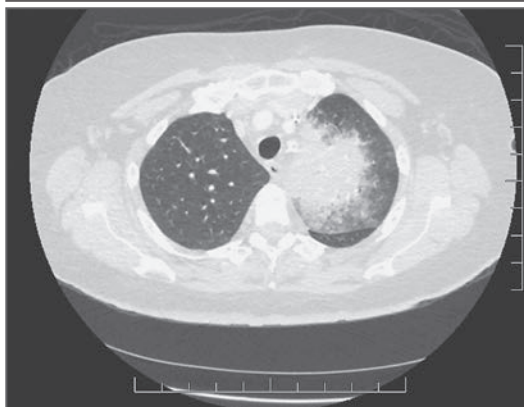


Figure 2. Computed Tomographic Scan of the Chest Obtained on Day 13 of Hospitalization.

A computed tomographic scan of the chest obtained on day 13 of hospitalization shows a masslike consolidative lesion in the left upper lobe with surrounding ground-glass opacification.

The area of consolidation in the left upper lobe probably represents a bacterial or fungal pneumonia. Cefepime is active against gram-negative pathogens and most causes of community-acquired pneumonia, but not methicillin-resistant *Staphylococcus aureus* or atypical bacteria such as legionella. Opportunistic bacterial infections such as those caused by nocardia and actinomyces can also manifest as pulmonary masses and require special culture conditions. Finally, despite prophylaxis with voriconazole, invasive fungal infections remain high on the differential diagnosis because of the risk factors associated with the host patient. The next step would be to obtain a sputum sample or perform a diagnostic bronchoscopy to obtain additional samples for microbiologic diagnosis.

The patient was unable to produce sputum for microbiologic testing. Urinary antigen tests for legionella and pneumococcus were negative. Serum beta-D-glucan and galactomannan levels were undetectable, and the serum voriconazole level was within the therapeutic range. A repeat nasopharyngeal test was negative for SARS-CoV-2. Cefepime therapy was continued, and intravenous therapies with vancomycin and tobramycin were initiated.

On hospital day 16, the patient became hypoxemic and was transferred to the intensive care unit to begin receiving supplemental oxygen through a heated high-flow nasal cannula, followed by emergency intubation. A bronchoscopy with bronchoalveolar lavage (BAL) was performed, and the leukocyte count in a BAL fluid sample was 161 per cubic millimeter, with 68% histiocytes, 22% lymphocytes, and 10% neutrophils. Gram stains, fungal calcofluor white stains, and acid-fast bacteria smears of alveolar fluid showed no organisms. Galactomannan levels were undetectable in a BAL fluid sample, and cytologic examination of the sample did not reveal malignant cells.

Negative microbiologic tests are not uncommon among patients with pneumonia, especially if they have received antecedent antibiotics. Negative urinary antigen tests for legionella and pneumococcus are insufficient to definitively rule out these pathogens. Urinary antigen tests for legionella detect *Legionella pneumophila* serogroup 1, which accounts for only 80% of cases of legionnaires' disease.

Most patients with acute respiratory tract in-

fections have BAL fluid differential cell counts that show predominant neutrophilia; the cellular profile here is atypical because the patient has neutropenia. At this point, my differential diagnosis would still include an infection such as legionella pneumonia that has not been fully treated. I also remain concerned about possible invasive fungal disease, recognizing that fungal biomarkers may be insensitive in patients receiving antifungal prophylaxis.

The patient became increasingly hypoxemic while receiving mechanical ventilation; chest radiographs showed progressive consolidation of the left lung fields and evolving air-space disease of the right lung (Fig. 3). Prophylaxis with voriconazole was discontinued, and therapy with liposomal amphotericin B was initiated.

The radiologic findings are somewhat atypical for invasive aspergillosis. Classic findings on computed tomography include cavitory lesions or nodules with a surrounding area of ground glass ("halo" sign). Endemic fungal infections such as histoplasmosis can mimic community-acquired pneumonia and merit consideration, given that the patient resides in Michigan and that voriconazole is not an adequate treatment for severe forms of these infections. Nocardiosis may produce pulmonary mass lesions and can

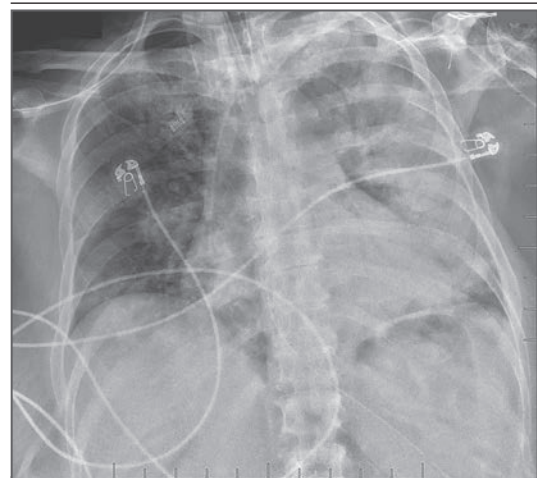


Figure 3. Chest Radiograph Obtained on Day 20 of Hospitalization.

A chest radiograph obtained on day 20 of hospitalization shows worsening consolidation of the left upper and lower lobes, with new air bronchograms, as well as evolving right lobar air-space disease.

progress to severe disease in immunocompromised hosts. Because of the clinical deterioration observed in the patient while she was receiving voriconazole, an escalation in therapy to liposomal amphotericin B is appropriate.

Legionnaires' disease, which is increasing in incidence in the United States, remains a possibility and should always be included in the differential diagnosis of severe pneumonia. PCR-based tests are now the method of choice for the diagnosis of legionella owing to their high sensitivity and rapid availability of results.

On hospital day 17, PCR tests of a BAL fluid sample were positive for *L. pneumophila* and *Pneumocystis jirovecii*. An electrocardiogram showed a corrected QT interval of 538 msec.

PCR testing for legionella can detect all serogroups of *L. pneumophila*, as well as species other than pneumophila. Nucleic acid amplification tests are the preferred method for the diagnosis of *P. jirovecii* infections; however, this sensitive assay cannot differentiate between infection and colonization by this ubiquitous organism. Of the two potential pathogens detected in the BAL samples from this patient, legionella is much more likely than pneumocystis to be the culprit. Pneumocystis infection is primarily associated with human immunodeficiency virus disease, long-term use of glucocorticoids, or other T-cell defects and is uncommon in patients with acute myeloid leukemia, unless they had received a hematopoietic stem-cell transplant. Moreover, the development of pulmonary consolidation with progression to acute respiratory distress syndrome in the patient is more consistent with legionella, whereas pneumocystis pneumonia typically manifests as diffuse interstitial infiltrates and often has a less rapidly progressive course. A serum beta-D-glucan test would be useful to potentially distinguish pneumocystis infection from colonization. At this point, I would initiate therapy with levofloxacin in the patient for coverage against legionella, given the risk of torsades de pointes with the use of macrolides. Although quinolones may also exacerbate prolongation of the QT interval, the clinical effect of this on adverse cardiac events is less clear than with macrolide therapy.

The patient began therapy with levofloxacin and doxycycline (macrolide therapy was not given be-

cause of the prolonged QT interval). All bronchoscopic cultures remained without growth, and the liposomal amphotericin B regimen was transitioned to isavuconazole. Treatment with high-dose trimethoprim-sulfamethoxazole for possible pneumocystis pneumonia was initiated but was quickly discontinued after a repeat serum beta-D-glucan test was negative. The patient recovered after nearly 2 weeks of therapy, and she was removed from the ventilator. She returned to the oncology ward for ongoing care for acute myeloid leukemia.

Bronchoscopic fluid was sent to the Michigan Department of Health and Human Services Bureau of Laboratories and grew *L. pneumophila* on buffered charcoal yeast extract media. PCR-based serotyping revealed serogroup 6. An epidemiologic investigation was performed in collaboration with the Michigan Department of Health and Human Services. In the month after the case of legionellosis was detected, multiple bulk water samples were collected on three separate occasions from the faucet and shower in the patient's room, the unit ice machine, and the proximal pressure tank. Water temperatures and monochloramine levels were within target ranges in all samples, and all cultures were negative for legionella growth. No further cases of legionellosis were noted in the hospital over the ensuing 3 months.

COMMENTARY

Legionella are gram-negative, facultative, intracellular bacteria with water and soil environmental reservoirs. Humans acquire legionella through inhalation of contaminated aerosols, and infection results in a spectrum of illness, including legionnaires' disease, a form of pneumonia (named after a famed outbreak at the 1976 Philadelphia Legionnaires' convention), as well as Pontiac fever, a nonspecific febrile illness observed predominantly in outbreak settings. The incidence of legionellosis is 6 times as high as it was two decades ago. This rise in incidence has occurred because of a confluence of factors including improved diagnostic capabilities, an aging infrastructure and population, and rising global temperatures resulting in a warmer, wetter climate.^{1,2}

Risk factors for legionella pneumonia include male sex, age greater than 50 years, cigarette smoking, impaired immunity, neuromuscular disease, and occupational exposure to engineered

water systems, such as cooling towers, hot tubs, showers, and decorative fountains.² The clinical and radiographic features of legionnaires' disease are often indistinguishable from pneumonia caused by other bacteria. Concurrent gastrointestinal symptoms (such as nausea and diarrhea) and selected laboratory findings (such as hyponatremia, hepatocellular liver injury, and elevated creatinine phosphokinase level) are frequently reported in case series of patients with legionella pneumonia, but these clinical features do not reliably discriminate legionella from other pathogens.³ Although legionella commonly presents with radiographic features analogous to other causes of bacterial pneumonia, imaging findings in immunocompromised patients may mimic invasive fungal infections — as was the case in the current patient — with masslike nodular lesions marked by a halo sign or cavitation.⁴ The initial conceptualization of the current case as a potential breakthrough invasive fungal infection that was based on the characteristics of the host and imaging findings probably contributed to anchoring bias (which occurs when clinicians settle on a diagnosis on the basis of initial information, despite contrary evidence) and precipitated delays in the diagnosis and treatment of legionellosis.

Because legionella grows poorly on routine culture media, diagnosis has historically relied on urinary antigen testing. This method detects *L. pneumophila* serogroup 1, the predominant cause of legionellosis, but not other *L. pneumophila* serogroups or the 30 other species of legionella that have been isolated from humans.² Urinary antigen testing consequently has a limited diagnostic sensitivity of only 75%,⁵ and the worldwide burden of legionnaires' disease is estimated to be up to 10 times as high as the number of cases reported.² PCR testing for legionella is an increasingly preferred diagnostic testing method that can detect all clinically relevant legionella serogroups and species. Premature diagnostic closure based on a negative urinary antigen test for legionella may have contributed to diagnostic delay in the current case.

The first-line treatment of legionella pneumonia is either azithromycin or levofloxacin, as determined primarily on the basis of observational data.⁶ Combination antimicrobial therapy is frequently used for severe disease, but data supporting efficacy are lacking.

Engineered water systems serve as legionella reservoirs. Factors that favor the proliferation of legionella include water temperatures between 25°C and 43°C, water stagnation, increased levels of organic matter, and plumbing biofilms. Turbulent water flow at distal sources facilitates transmission to humans through contaminated aerosols. Health care facilities are notable potential legionella reservoirs because of their complex plumbing formations and vulnerable populations with increased susceptibility to legionellosis; an estimated 20% of reported cases in the United States are nosocomially acquired.⁷ It is therefore critical for clinicians to consider legionella as a potential cause of hospital-acquired pneumonia, particularly in patients who do not have a response to beta-lactam monotherapy.

It is unclear whether the legionella infection in the current patient was nosocomially acquired. The established incubation period for legionella is 2 to 10 days, and the patient's fever began on day 10 of hospitalization. In addition, an investigation did not detect nosocomial reservoirs of legionella, and no further clinical cases were identified in the following months. Although rare, prolonged incubation periods between exposure and disease have been reported for legionella in immunocompromised patients with community exposure,² and community exposure is plausible in relation to this patient's janitorial work at a health care facility or travel to a casino before hospitalization (although no further cases of legionella infection at these locations were reported).^{8,9}

The possibility of community acquisition of legionella is of particular relevance in the state of Michigan, which has had an increase in the incidence of legionellosis over the past decade.¹⁰ The Flint water crisis, which involved the occurrence of numerous cases of lead poisoning after the source and treatment of public water for Flint, Michigan, were changed as a cost-saving measure in April 2014, was coincident with large outbreaks of legionnaires' disease. A total of 79 cases of legionellosis, including 12 cases that resulted in death, occurred between 2014 and 2015; during this period, the risk of legionella infection among the residents of Flint was 6.3 times as high as that among historical controls.¹¹ Available evidence suggests that the switch to a more corrosive water source without appropriate safeguards in water resource management resulted

in reduced levels of residual chlorine, disturbance of biofilms, and accumulation of free iron and organic matter, which created a favorable environment for legionella growth.^{2,11,12} Our patient was infected with *L. pneumophila* serogroup 6, the predominant type in residences in Flint¹³ and in portions of Ontario,¹⁴ a Canadian province adjacent to southeast Michigan. Whether environmental conditions in these regions are especially conducive to serogroup 6 strains has not been investigated. The Flint water crisis and other legionella outbreaks highlight the critical but often overlooked role of social determinants of health in clinical decision making. Census-level data show that the risk of legionellosis correlates

geographically with areas with higher rates of poverty, in which non-White populations are disproportionately represented.¹⁵

Owing to a changing climate and aging water distribution systems with resultant increased corrosion and biofilm formation, waterborne pathogens such as legionella represent increasingly common challenges to clinicians. This case highlights the difficulties in making such a fluid diagnosis.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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REFERENCES

1. Beauté J, European Legionnaires' Disease Surveillance Network. Legionnaires' disease in Europe, 2011 to 2015. *Euro Surveill* 2017;22:30566.
2. National Academies of Sciences, Engineering, and Medicine. Management of legionella in water systems. Washington, DC: National Academies Press, 2020.
3. Sopena N, Sabrià-Leal M, Pedro-Boté ML, et al. Comparative study of the clinical presentation of *Legionella pneumoniae* and other community-acquired pneumonias. *Chest* 1998;113:1195-200.
4. Muder RR, Yu VL, Parry MF. The radiologic manifestations of *Legionella pneumoniae*. *Semin Respir Infect* 1987;2:242-54.
5. Shimada T, Noguchi Y, Jackson JL, et al. Systematic review and metaanalysis: urinary antigen tests for Legionellosis. *Chest* 2009;136:1576-85.
6. Jasper AS, Musuuza JS, Tischendorf JS, et al. Are fluoroquinolones or macrolides better for treating *Legionella pneumoniae*? A systemic review and meta-analysis. *Clin Infect Dis* 2021;72:1979-89.
7. Soda EA, Barskey AE, Shah PP, et al. Health care-associated Legionnaires' disease surveillance data from 20 states and a large metropolitan area — United States, 2015. *MMWR Morb Mortal Wkly Rep* 2017;66:584-9.
8. Bargellini A, Marchesi I, Marchegiano P, et al. A culture-proven case of community-acquired legionella pneumonia apparently classified as nosocomial: diagnostic and public health implications. *Case Rep Med* 2013;2013:303712.
9. Den Boer JW, Yzerman EPF, Schellekens J, et al. A large outbreak of Legionnaires' disease at a flower show, the Netherlands, 1999. *Emerg Infect Dis* 2002;8:37-43.
10. Michigan Department of Health & Human Services. Michigan experiencing increase in legionellosis cases. July 5, 2019 (<https://www.michigan.gov/mdhhs/0,5885,7-339--501322--,00.html>).
11. Zahran S, McElmurry SP, Kilgore PE, et al. Assessment of the Legionnaires' disease outbreak in Flint, Michigan. *Proc Natl Acad Sci U S A* 2018;115(8):E1730-E1739.
12. Rhoads WJ, Garner E, Ji P, et al. Distribution system operational deficiencies coincide with reported Legionnaires' disease clusters in Flint, Michigan. *Environ Sci Technol* 2017;51:11986-95.
13. Byrne BG, McColm S, McElmurry SP, et al. Prevalence of infection — competent serogroup 6 *Legionella pneumophila* within premise plumbing in southeast Michigan. *mBio* 2018;9(1):e00016-18.
14. Dutka BJ, Walsh K, Ewan P, El-Shaarawi A, Tobin RS. Incidence of *Legionella* organisms in selected Ontario (Canada) cities. *Sci Total Environ* 1984;39:237-49.
15. Gleason JA, Ross KM, Greeley RD. Analysis of population-level determinants of legionellosis: spatial and geovisual methods for enhancing classification of high-risk areas. *Int J Health Geogr* 2017;16:45.

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